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广州市耀国技术服务有限公司
检测专用章

REPORT ON THE SURFACE SPREAD OF FLAME TEST TO BS 476 : PART 7 : 1997 ON "Polyurethane Foam (PU) of Double Skin Panel"

Client	:	Dongguan Tech Free Air Conditioning Mfy. Ltd.
Project	:	Fire Test on Building Materials & Structures Part 7
Client Ref.	:	--
Report No.	:	G20849FU206109
Date	:	08 Jul 2020

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1.0 Introduction

Fugro Technical Services (Guangzhou) Limited was commissioned by the client, Dongguan Tech Free Air Conditioning Mfy. Ltd. to conduct a surface spread of flame test of Polyurethane Foam (PU) of Double Skin Panel in accordance with **BS 476 : Part 7 : 1997 Incorporating Corrigendum** No.1:2014. The test was conducted by Fugro Technical Services (Guangzhou) Limited's approved sub-contract Laboratory.

The test samples were prepared by the client and received by Fugro Technical Services (Guangzhou) Limited on 11 June 2020 and given Lab. Code : FU206109.

2.0 Description of Samples

Six pieces of specimen, said to be "Polyurethane Foam (PU) of Double Skin Panel", each of nominal size of 885 mm x 270 mm were submitted. The overall bulk density and thickness of the specimen was found to be approximately 45±2 kg/m^3 and 50 mm respectively.

Name and address of the sponsor	Dongguan Tech Free Air Conditioning Mfy. Ltd. No. 11 Daji Industrial Zone, Shipai Town, Dongguan City, Guangdong Province, P.R.C.
Name and address of the manufacturer/supplier	Dongguan Tech Free Air Conditioning Mfy. Ltd. No. 11 Daji Industrial Zone, Shipai Town, Dongguan City, Guangdong Province, P.R.C.
General description	Polyurethane Foam (PU) of Double Skin Panel
Material & composition for composite product	Aluminium Metal Skin, PU Inter-layer
Nominal thickness	50 mm
Nominal density	45±2 kg/m^3
Nominal mass per unit area	Nil
Colour	Silvery
Brand Name	TECH FREE
Model	RH2038101(P50AL)

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3.0 Purpose of Test

To determine the tendency of the surface of a material or a combination of materials to support the spread of flame across its surface and to classify the surface according to the test given in British Standard 476 : Part 7 : 1997 Incorporatig Corrigendum No.1:2014.

4.0 Test Procedure

Prior to test, the specimens were prepared and conditioned in accordance with paragraphs 5.3 to 5.6 of the standard and secured to a specimen holder as described in paragraph 6.3.

Six specimens were tested with the smooth face exposed to the specified thermal radiation from the apparatus described in paragraph 6.1 of the standard.
The intensity of the radiated heat incident on the specimen varies with distance from the hotter end, so that when the specified calibration panel is mounted in the place to be occupied by the specimen, the irradiance of the radiometer is as given in Table 1.
The test was terminated when the flame front reached the 825 mm reference line, or after 10 minutes has elapsed, whichever is the shorter.

Table 1 : Irradiance Along Horizontal Reference Line on the Calibration Board

Distance along reference line from inside edge of specimen holder (mm)	Irradiance (kW/m^2)		
	specified	min.	max.
75	32.5	32.0	33.0
225	21.0	20.5	21.5
375	14.5	14.0	15.0
525	10.0	9.5	10.5
675	7.0	6.5	7.5
825	5.0	4.5	5.5

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5.0 Test Performing Date

22 June 2020 to 01 July 2020

6.0 Results of Test

Specimen No.	1	2	3	4	5	6
Distance (mm)	Time of spread of flame to indicated distance (minutes • seconds)					
75	--	--	--	--	--	--
165	--	--	--	--	--	--
190	--	--	--	--	--	--
215	--	--	--	--	--	--
240	--	--	--	--	--	--
265	--	--	--	--	--	--
290	--	--	--	--	--	--
375	--	--	--	--	--	--
455	--	--	--	--	--	--
500	--	--	--	--	--	--
525	--	--	--	--	--	--
600	--	--	--	--	--	--
675	--	--	--	--	--	--
710	--	--	--	--	--	--
750	--	--	--	--	--	--
785	--	--	--	--	--	--
825	--	--	--	--	--	--
865	--	--	--	--	--	--
Maximum distance traveled at 1.5 minutes (mm)	<50	<50	<50	<50	<50	<50
Maximum distance traveled during the whole test (mm)	--	--	--	--	--	--
Time to reach maximum distance traveled	--	--	--	--	--	--
Comments	Observations during the test: The specimens did not ignite.					

Note:

1. "--" Not reached the reference line
2. Six specimens are usually tested. If the test on any specimen is deemed to be invalid, as defined in the standard, it is permissible for up to a maximum of nine specimens to be tested in order to obtain the six valid test results.

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7.0 Conclusion

In accordance with the class definitions specified in the Standard, the test results show that the sample tested has a **Class 1 Surface Spread of Flame**.

Classification of Surface Spread of Flame

Classification	Spread of flame at 1.5 min.		Final spread of flame	
	Limit (mm)	Limit for one specimen in sample (mm)	Limit (mm)	Limit for one specimen in sample (mm)
Class 1	165	165 + 25	165	165 + 25
Class 2	215	215 + 25	455	455 + 45
Class 3	265	265 + 25	710	710 + 75
Class 4	Exceeding the limits for class 3			

Explanation of prefix and suffixes which may be added to the classification

1. A suffix R is added to the classification if more than six specimens are required in order to obtain six valid test results (e.g. class 2R).
2. A prefix D is added to the classification of any product which does not comform to the surface characteristics specified in the standard and has therefore been tested in a modified form (e.g. class D3).
3. A suffix Y shall be added to the classification if any softening and/or other behaviour that may affect the flame spread occurs.

Statement: The test results relate only to the behaviour of the test specimens of the product under the particular conditions of test ; they are not intended to be the sole criterion for assessing the potential fire hazard of the product in use.

Checked by : Eden

Approved by :
Yan Cai Sheng (Deputy General Manager)

Date : 08/07/2020

Date : 08/07/2020

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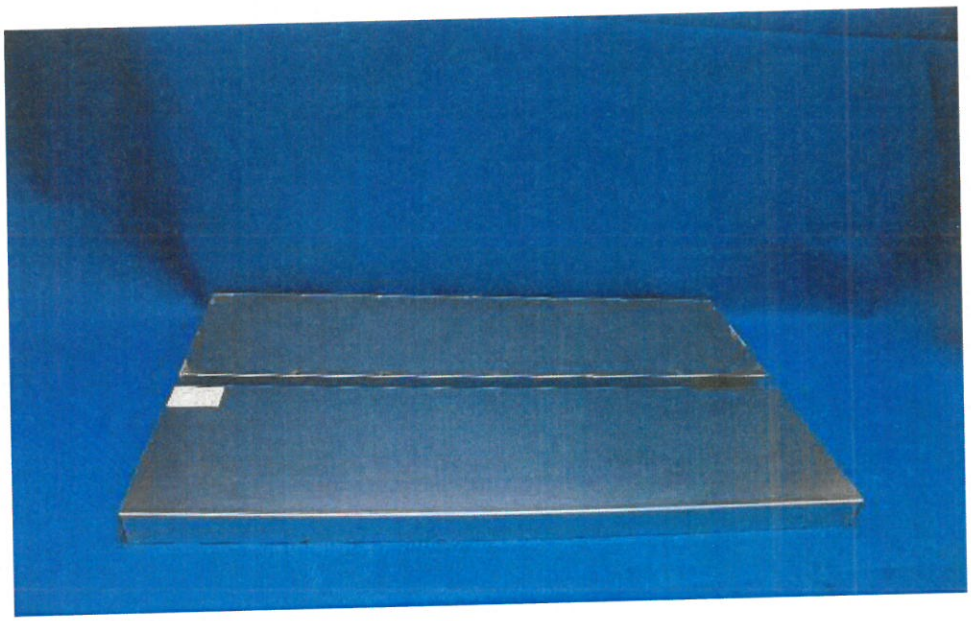
Appendix **Specimen Photo**



Sample ID: FU206109

** End of Report **

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